

Lecture 7: Confidence intervals

Economics 326 — Methods of Empirical Research in Economics

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Point vs interval estimators

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- Recall our model:

- $Y_i = \beta_0 + \beta_1 X_i + U_i, \quad i = 1, \dots, n.$
- $E[U_i | \mathbf{X}] = 0$ for all i 's.
- $E[U_i^2 | \mathbf{X}] = \sigma^2$ for all i 's.
- $E[U_i U_j | \mathbf{X}] = 0$ for all $i \neq j$.

- So far we have established that conditionally on $\mathbf{X}$:

- $E[\hat{\beta}_1 | \mathbf{X}] = \beta_1$ (unbiasedness),
- $\text{Var}(\hat{\beta}_1 | \mathbf{X}) = \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}.$

- If the U_i 's are continuously distributed, then with probability **one**, $\hat{\beta}_1 \neq \beta_1$:

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^n (X_i - \bar{X}) U_i}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

- An **interval estimator** is a random interval $[LB, UB]$ that contains the true parameter value with a pre-specified probability.
- To construct an interval estimator for β_1 , we need to know the **distribution** of $\hat{\beta}_1$.
- This requires an additional assumption about the distribution of U_i 's. Let's first review the normal distribution.

Normal distribution

- A normal rv is a continuous rv that can take on any value. The PDF of a normal rv X is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \text{ where}$$

$$\mu = E[X] \text{ and } \sigma^2 = \text{Var}(X).$$

We usually write $X \sim N(\mu, \sigma^2)$.

- If $X \sim N(\mu, \sigma^2)$, then $a + bX \sim N(a + b\mu, b^2\sigma^2)$.

Standard normal distribution

- A standard normal rv has $\mu = 0$ and $\sigma^2 = 1$. Its PDF is $\phi(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right)$.
- Symmetric around zero (mean): if $Z \sim N(0, 1)$,

$$P(Z > 0) = P(Z < 0) = 0.5,$$

$$P(Z > z) = P(Z < -z) \text{ for any } z.$$

- Thin tails: $P(-1.96 \leq Z \leq 1.96) = 0.95$.
- If $X \sim N(\mu, \sigma^2)$, then $(X - \mu)/\sigma \sim N(0, 1)$.

Bivariate normal distribution

- X and Y have a bivariate normal distribution if their joint PDF is given by:

$$f(x, y) = \frac{1}{2\pi\sqrt{(1-\rho^2)\sigma_X^2\sigma_Y^2}} \exp\left[-\frac{Q}{2(1-\rho^2)}\right],$$

where

$$Q = \frac{(x - \mu_X)^2}{\sigma_X^2} + \frac{(y - \mu_Y)^2}{\sigma_Y^2} - 2\rho \frac{(x - \mu_X)(y - \mu_Y)}{\sigma_X\sigma_Y},$$

$\mu_X = E[X]$, $\mu_Y = E[Y]$, $\sigma_X^2 = \text{Var}(X)$, $\sigma_Y^2 = \text{Var}(Y)$, and $\rho = \text{Corr}(X, Y)$.

Properties of bivariate normal

- If X and Y have a bivariate normal distribution, then $a + bX + cY \sim N(\mu^*, (\sigma^*)^2)$, where

$$\begin{aligned}\mu^* &= a + b\mu_X + c\mu_Y, \\ (\sigma^*)^2 &= b^2\sigma_X^2 + c^2\sigma_Y^2 + 2bc\rho\sigma_X\sigma_Y.\end{aligned}$$

- $\text{Cov}(X, Y) = 0 \implies X$ and Y are independent.
- This can be generalized to more than 2 variables (multivariate normal).

Normality of the OLS estimator

- **Assumption 5:** U 's are jointly normally distributed conditional on $\mathbf{X}$.
- Then $Y_i = \beta_0 + \beta_1 X_i + U_i$ are also jointly normally distributed conditional on $\mathbf{X}$.
- Since $\hat{\beta}_1 = \sum_{i=1}^n w_i Y_i$, where $w_i = \frac{X_i - \bar{X}}{\sum_{i=1}^n (X_i - \bar{X})^2}$ depend only on $\mathbf{X}$, $\hat{\beta}_1$ is also normally distributed conditional on $\mathbf{X}$.
- Conditionally on $\mathbf{X}$:

$$\begin{aligned}\hat{\beta}_1 | \mathbf{X} &\sim N(\beta_1, \text{Var}(\hat{\beta}_1 | \mathbf{X})), \\ \text{Var}(\hat{\beta}_1 | \mathbf{X}) &= \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}.\end{aligned}$$

Interval estimation problem

- We want to construct an **interval estimator** for β_1 :
 - The interval estimator is called a **confidence interval** (CI).
 - A CI contains the **true** value β_1 **with some pre-specified probability** $1 - \alpha$, where α is a small probability of error.
 - For example, if $\alpha = 0.05$, then the random CI will contain β_1 with probability 0.95.
- $1 - \alpha$ is called the **coverage probability**.
- Confidence interval: $CI_{1-\alpha} = [LB_{1-\alpha}, UB_{1-\alpha}]$. The lower bound (LB) and upper bound (UB) should depend on the coverage probability $1 - \alpha$.

- The formal definition of CI: It is a **random interval** $CI_{1-\alpha}$ such that conditionally on $\mathbf{X}$,

$$P(\beta_1 \in CI_{1-\alpha} \mid \mathbf{X}) = 1 - \alpha.$$

Note that the random element is $CI_{1-\alpha}$.

- Sometimes, a CI is defined as $P(\beta_1 \in CI_{1-\alpha}) \geq 1 - \alpha$.

Symmetric CIs

- One approach to constructing CIs is to consider a **symmetric** interval around the estimator $\hat{\beta}_1$:

$$CI_{1-\alpha} = [\hat{\beta}_1 - c_{1-\alpha}, \hat{\beta}_1 + c_{1-\alpha}]$$

- The problem is choosing $c_{1-\alpha}$ such that $P(\beta_1 \in CI_{1-\alpha} \mid \mathbf{X}) = 1 - \alpha$.
- In choosing $c_{1-\alpha}$, we will be relying on the fact that, given our assumptions and conditionally on $\mathbf{X}$:

$$\begin{aligned} \hat{\beta}_1 \mid \mathbf{X} &\sim N(\beta_1, \text{Var}(\hat{\beta}_1 \mid \mathbf{X})), \\ \text{Var}(\hat{\beta}_1 \mid \mathbf{X}) &= \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}. \end{aligned}$$

- Note that conditionally on $\mathbf{X}$:

$$\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\text{Var}(\hat{\beta}_1 \mid \mathbf{X})}} \sim N(0, 1).$$

Standard normal quantiles

- Let $Z \sim N(0, 1)$. The τ -th **quantile** (percentile) of the standard normal distribution is z_τ such that

$$P(Z \leq z_\tau) = \tau.$$

- **Median:** $\tau = 0.5$ and $z_{0.5} = 0$. ($P(Z \leq 0) = 0.5$).
- If $\tau = 0.975$ then $z_{0.975} = 1.96$. Due to symmetry, if $\tau = 0.025$ then $z_{0.025} = -1.96$.

σ^2 is known (infeasible CIs)

- **Suppose** (for a moment) that σ^2 is known, and we can compute exactly the variance of $\hat{\beta}_1$:

$$\text{Var}(\hat{\beta}_1 \mid \mathbf{X}) = \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

- Consider the following CI:

$$CI_{1-\alpha} = \left[\hat{\beta}_1 - z_{1-\alpha/2} \sqrt{\text{Var}(\hat{\beta}_1 \mid \mathbf{X})}, \hat{\beta}_1 + z_{1-\alpha/2} \sqrt{\text{Var}(\hat{\beta}_1 \mid \mathbf{X})} \right].$$

- For example, if $1 - \alpha = 0.95 \iff \alpha = 0.05 \iff z_{1-\alpha/2} = z_{0.975} = 1.96$, and $CI_{0.95}$ is

$$\hat{\beta}_1 \pm 1.96 \sqrt{\text{Var}(\hat{\beta}_1 \mid \mathbf{X})}.$$

Infeasible CI validity (σ^2 known)

- **Goal:** show that $P(\beta_1 \in CI_{1-\alpha} \mid \mathbf{X}) = 1 - \alpha$.
- **Notation:** $\sigma_{\hat{\beta}_1} = \sqrt{\text{Var}(\hat{\beta}_1 \mid \mathbf{X})}$.
- **Key fact:** $Z = \frac{\hat{\beta}_1 - \beta_1}{\sigma_{\hat{\beta}_1}} \sim N(0, 1)$ conditionally on $\mathbf{X}$.

$$\begin{aligned}
& P(\beta_1 \in CI_{1-\alpha} \mid \mathbf{X}) \\
&= P(\hat{\beta}_1 - z_{1-\alpha/2} \sigma_{\hat{\beta}_1} \leq \beta_1 \leq \hat{\beta}_1 + z_{1-\alpha/2} \sigma_{\hat{\beta}_1} \mid \mathbf{X}) \\
&= P(-z_{1-\alpha/2} \sigma_{\hat{\beta}_1} \leq \beta_1 - \hat{\beta}_1 \leq z_{1-\alpha/2} \sigma_{\hat{\beta}_1} \mid \mathbf{X}) \\
&= P(-z_{1-\alpha/2} \sigma_{\hat{\beta}_1} \leq \hat{\beta}_1 - \beta_1 \leq z_{1-\alpha/2} \sigma_{\hat{\beta}_1} \mid \mathbf{X}) \\
&= P\left(-z_{1-\alpha/2} \leq \frac{\hat{\beta}_1 - \beta_1}{\sigma_{\hat{\beta}_1}} \leq z_{1-\alpha/2} \mid \mathbf{X}\right) \\
&= P(-z_{1-\alpha/2} \leq Z \leq z_{1-\alpha/2} \mid \mathbf{X}) \\
&= P(Z \leq z_{1-\alpha/2} \mid \mathbf{X}) - P(Z \leq -z_{1-\alpha/2} \mid \mathbf{X}) \\
&= (1 - \alpha/2) - \alpha/2 \\
&= 1 - \alpha.
\end{aligned}$$

Feasible CIs (σ^2 unknown)

- Since σ^2 is unknown, we must estimate it from the data:

$$\begin{aligned}
s^2 &= \frac{1}{n-2} \sum_{i=1}^n \hat{U}_i^2 \\
&= \frac{1}{n-2} \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2.
\end{aligned}$$

- The **standard error** of $\hat{\beta}_1$ is defined as

$$\begin{aligned}
\text{se}(\hat{\beta}_1) &= \sqrt{\widehat{\text{Var}}(\hat{\beta}_1)} \\
&= \sqrt{\frac{s^2}{\sum_{i=1}^n (X_i - \bar{X})^2}}.
\end{aligned}$$

- Replacing σ by its estimate does not give a normal distribution anymore:

$$\frac{\hat{\beta}_1 - \beta_1}{\text{se}(\hat{\beta}_1)} \mid \mathbf{X} \sim t_{n-2}.$$

Here t_{n-2} denotes the t -distribution with $n - 2$ degrees of freedom.

- The degrees of freedom depend on
 - the sample size (n),
 - and the number of parameters one has to estimate to compute s^2 (two in this case, β_0 and β_1).

Feasible CIs (σ^2 unknown)

- Let $t_{df,\tau}$ be the τ -th quantile of the t -distribution with the number of degrees of freedom df : If $T \sim t_{df}$ then

$$P(T \leq t_{df,\tau}) = \tau.$$

- Similarly to the normal distribution, the t -distribution is centered at zero and is symmetric around zero:
 $t_{n-2, 1-\alpha/2} = -t_{n-2, \alpha/2}$.
- We can now construct a feasible $CI_{1-\alpha}$ as

$$\hat{\beta}_1 \pm t_{n-2, 1-\alpha/2} \times \text{se}(\hat{\beta}_1).$$

Example

- Data: **rental** from the **wooldridge** R package. 64 US cities in 1990.
 - **rent**: average monthly rent (\$)
 - **avginc**: per capita income (\$)
- Model: $\text{Rent}_i = \beta_0 + \beta_1 \text{AvgInc}_i + U_i$.
- R implementation:

```
# Load data and run OLS regression
library(wooldridge)
data("rental")
rental90 <- subset(rental, y90 == 1)
reg <- lm(rent ~ avginc, data = rental90)
summary(reg)
```

Call:

```
lm(formula = rent ~ avginc, data = rental90)
```

Residuals:

Min	1Q	Median	3Q	Max
-94.67	-47.27	-13.68	25.65	228.46

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.488e+02	3.210e+01	4.635	1.89e-05 ***
avginc	1.158e-02	1.308e-03	8.851	1.34e-12 ***

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 66.56 on 62 degrees of freedom
 Multiple R-squared: 0.5582, Adjusted R-squared: 0.5511
 F-statistic: 78.34 on 1 and 62 DF, p-value: 1.341e-12

- 95% CI for the slope coefficient:

```
confint(reg, "avginc", level = 0.95)
```

	2.5 %	97.5 %
avginc	0.008964625	0.01419539

- 90% CI for the slope coefficient:

```
confint(reg, "avginc", level = 0.90)
```

	5 %	95 %
avginc	0.009395296	0.01376472

The effect of estimating σ^2

- The t -distribution has heavier tails than the normal.
- $t_{df, 1-\alpha/2} > z_{1-\alpha/2}$, but as df increases $t_{df, 1-\alpha/2} \rightarrow z_{1-\alpha/2}$.

- When the sample size n is large, $t_{n-2, 1-\alpha/2}$ can be replaced with $z_{1-\alpha/2}$.
- In R, use `qt()` for t -quantiles and `qnorm()` for z -quantiles:

```
# z critical value for 95% CI
qnorm(0.975)
```

```
[1] 1.959964
```

```
# t critical values for 95% CI with different df
qt(0.975, df = 30)
```

```
[1] 2.042272
```

```
qt(0.975, df = 100)
```

```
[1] 1.983972
```

```
qt(0.975, df = 1000)
```

```
[1] 1.962339
```

```
qt(0.975, df = 10000)
```

```
[1] 1.960201
```

Interpretation of confidence intervals

- The confidence interval $CI_{1-\alpha}$ is a function of the **sample** $\{(Y_i, X_i) : i = 1, \dots, n\}$, and therefore is **random**. This allows us to talk about the probability of $CI_{1-\alpha}$ containing the true value of β_1 .
- Once the confidence interval is computed given the data, we have its **one realization**. The realization of $CI_{1-\alpha}$ (the computed confidence interval) is not random, and it does not make sense anymore to talk about the probability that it includes the true β_1 .
- **Once the confidence interval is computed, it either contains the true value β_1 or it does not.**